Information Security

Lecture 7

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Announcements:

Assignment 1 is due soon!

There will be a test on Monday, March 16
on the material shared before the break.

What is a Natural Number?

1. Zero is a natural number.
2. Equality is reflexive, symmetric, and transitive; and natural numbers are closed under equality.
3. For every natural number there is a successor.
4. Successor is an injective function.
5. Zero is not the successor of any function.
6. Induction!

$\mathbb{N}$

What is an Integer?

An integer is the difference between two natural numbers.

Every integer can be thought of as a set of pairs:

$$-3 = \{(8,11), (32,35), (2,5), \ldots\}$$

$$\mathbb{Z} \cong (\mathbb{N} \times \mathbb{N})/\{((a,b),(c,d)) \mid a + d = b + c\}$$

$$\mathbb{Z}$$

Modular Arithmetic

$$k\mathbb{Z} = \{\, kz \mid z \in \mathbb{Z} \,\}$$

$$\mathbb{Z}/k\mathbb{Z} = \{\, a + k\mathbb{Z} \mid a \in \mathbb{Z} \,\}$$

$$\mathbb{Z}$$

That is, $a \equiv b \mod k \iff [a - b] \in k\mathbb{Z}$

$$\iff \exists z \in \mathbb{Z} \,.\, (a - b) = kz$$

Modular Arithmetic

Example:

Let $k = 5$. Let $[2] \in \mathbb{Z}/5\mathbb{Z}$.

Then $[2] = \{2 + 5z : z \in \mathbb{Z}\} = \{\ldots, -8, -3, 2, 7, 12, \ldots\}$

Where... $-3 = \{(8,11), (32,35), (2,5), \ldots\}$

$\mathbb{Z}$

Modular Arithmetic

Let $a, b \in \mathbb{Z}/k\mathbb{Z}$, then using these defintions, check that

$$[a] + [b] = [a + b]$$

$$n[a] = [na]$$

$$[a]^n = [a^n]$$

What about the following:

$$[a] \times [b] = \{\alpha \times \beta : \alpha \in [a], \beta \in [b]\} = [a \times b]$$

Modular Arithmetic

If $a \equiv \alpha \mod k$ and $b \equiv \beta \mod k$,

then prove that $a \times b = \alpha \times \beta$.

Modular Arithmetic

If $a \equiv \alpha \mod k$ and $b \equiv \beta \mod k$,
then prove that $a \times b = \alpha \times \beta$.

By definition, there exists z and z'
such that:

$$a - \alpha = kz \quad b - \beta = kz'$$

Then:

$$ab = (\alpha + kz)(\beta + kz') = \alpha\beta + z(_)$$

Modular Arithmetic

Let $a, b \in \mathbb{Z}/k\mathbb{Z}$, then using these defintions, check that

$$[a] + [b] = [a + b]$$

$$[a] \times n = [a \times n]$$

$$[a]^n = [a^n]$$

$$[a] \times [b] = [a \times b]$$

What about the following:

$$[a]^{[b]} = [a^b]?$$

This will be in the test!

Groups, Rings, and Fields

An algebraic structure $(R, +, \times, 0, 1)$ is a ring if $(R, +, 0)$ is an Abelian group, $(R, \times, 1)$ is a monoid and $\times$ distributes over $+$.

$\mathbb{Z}/k\mathbb{Z}$ forms a ring.

If $R \backslash \{0\}$ has inverses under $\times$, we have a field.

Groups, Rings, and Fields

Problem 1: When is $\mathbb{Z}/k\mathbb{Z}$ a field?

That for every $x \in \mathbb{Z}/k\mathbb{Z} \backslash \{[0]\}$, we need an a such that $[ax] = [1]$.

Groups, Rings, and Fields

Problem 1: When is $\mathbb{Z}/k\mathbb{Z}$ a field?

That for every $x \in \mathbb{Z}/k\mathbb{Z}\backslash\{0\}$, we need an a such that $ax \equiv 1 \mod k$.

That is, we need a and b such that

$$ax + bk = 1.$$

Groups, Rings, and Fields

Fixed x and k, when do we have a and b such that $ax + bk = 1$?

When $\gcd(x, k) = 1$.

Complexity?

Groups, Rings, and Fields

Fixed x and k, when do we have a and b such that $ax + bk = 1$?

When $\gcd(x, k) = 1$.

Let $\left(\mathbb{Z}/k\mathbb{Z}\right)^* = \{[x] : \gcd(x, k) = 1\}$

It forms a group under multiplication!

Groups, Rings, and Fields

When k is prime, for all $x \not\equiv 0 \mod k$,
we have hen $\gcd(x, k) = 1$.

$$\left(\mathbb{Z}/p\mathbb{Z}\right)^* = \{[1], [2], \ldots, [p-1]\}$$

Modular Linear Equations

When is there a solution to

$$ax \equiv b \mod k?$$

If and only if $a \in \left(\mathbb{Z}/k\mathbb{Z}\right)^*$.

How many elements are there in $\left(\mathbb{Z}/k\mathbb{Z}\right)^*$?

Given n, what is the cardinality of the set:

$$S = \{x : x \in \mathbb{N}, x < n, \gcd(x, n) = 1\}$$

If n is prime, then $|S| = p - 1$

How many elements are there in $\left(\mathbb{Z}/k\mathbb{Z}\right)^*$?

If $n = p^2$ for some prime p, then?

$$S = \{1,2,\ldots,p-1,p+1,p+2,\ldots 2p-1,2p+1,\ldots p^2-1\}$$

$$\varphi(p^2) = p(p-1)$$

How many elements are there in $\left(\mathbb{Z}/k\mathbb{Z}\right)^*$?

For co-prime a and b, prove that

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Fermat's Little Theorem

For a prime number p, if $p \nmid a$, then:

$$a^{p-1} \equiv 1 \mod p.$$

Fermat's Little Theorem

Proof: Consider $\{1,2,\ldots,(p-1)\}$.

Multiply by a, and consider them $\bmod p$:

$$\{a,2a,\ldots,(p-1)a\}.$$

None of them are 0, and each one is distinct.

Fermat's Little Theorem

Therefore,

$$1 \times 2 \times \cdots \times (p-1) \equiv a \times 2a \times \cdots \times a(p-1) \mod p$$

$$\equiv a^{p-1}(p-1)! \mod p$$

Fermat's Little Theorem

For a prime number p, if $p \nmid a$, then:

$$a^{p-1} \equiv 1 \mod p.$$

if $p \mid a$, then: $a^{p-1} \equiv 0 \mod p.$

Therefore, $a^p \equiv a \mod p.$

Euler's Theorem

If $a \in \left(\mathbb{Z}/k\mathbb{Z}\right)^*$, $a^{\varphi(k)} \equiv 1 \mod k$.

Proof: Consider $\{x_1, x_2, \ldots, x_n\} = \left(\mathbb{Z}/k\mathbb{Z}\right)^*$.

Each ax_i is unique and non-zero. Hence:

$$\prod x_i \equiv a^{\varphi(k)} \prod x_i \mod k$$

How does this connect to Information Security?

Symmetric Key Encryption

Asymmetric Key Encryption

Diffie-Hellman key exchange protocol

RSA encryption

Symmetric Key Encryption

How to securely share the secret key among each pair of communicating parties?

Suppose there is a protocol for A to send the key $k_{A,B}$ to B, then it can use the same protocol to also exchange messages!

Diffie-Hellman key exchange protocol

Protocol begins by choosing a publicly agreed upon a large prime p and the associated primitive root $g \in (\mathbb{Z}/p\mathbb{Z})^*$.

Primitive Generator

$$\left(\mathbb{Z}/p\mathbb{Z}\right)^* = \{[1], [2], \ldots, [p-1]\}$$

There exists $g \in (\mathbb{Z}/p\mathbb{Z})^*$ such that

$$\{g, g^2, \ldots, g^{p-1}\} = (\mathbb{Z}/p\mathbb{Z})^*$$

Proof?

Diffie-Hellman key exchange protocol

Protocol begins by choosing a publicly agreed upon a large prime p and an associated primitive root $g \in (\mathbb{Z}/p\mathbb{Z})^*$.

Two participants A and B choose keys a and b separately.

A computes $g^a \mod p$, and sends it to B.

B computes $g^b \mod p$, and sends it to A.

Diffie-Hellman key exchange protocol

Protocol begins by choosing a publicly agreed upon a large prime p and an associated primitive root $g \in (\mathbb{Z}/p\mathbb{Z})^*$.

Two participants A and B choose keys a and b separately.

A uses the recieved number to compute $(g^b)^a \mod p$.

B uses the recieved number to compute $(g^a)^b \mod p$.

Diffie-Hellman key exchange protocol

Protocol begins by choosing a publicly agreed upon a large prime p and an associated primitive root $g \in (\mathbb{Z}/p\mathbb{Z})^*$.

The shared key is g^{ab}.

How will you compute it? What are the assumptions?

Diffie-Hellman key exchange protocol

How will you compute it? What are the assumptions?

Disadvantages:

1. For each agent B in the system, A must maintain a separate key.
2. There is a MITM attack!!

Diffie-Hellman key exchange protocol

$$A \rightarrow I : g^a$$

$$I \rightarrow B : g^x \qquad k_{A,I} = g^{ay}$$

$$B \rightarrow I : g^b \qquad k_{B,I} = g^{bx}$$

$$I \rightarrow A : g^y$$

Intruder can keep communicating with both A and B.

This can be resolved by authentication.

But does that make it secure?

en.wikipedia.org/wiki/Logjam_(computer_security)

Imperfect Forward Secrecy: How Diffie-Hellman Fails in Practice

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For additional materials and contact information, visit WeakDH.org.

ABSTRACT

We investigate the security of Diffie-Hellman key exchange as used in popular Internet protocols and find it to be less secure than widely believed. First, we present Logjam, a novel flaw in TLS that lets a man-in-the-middle downgrade connections to "export-grade" Diffie-Hellman. To carry out this attack, we implement the number field sieve discrete log algorithm. After a week-long precomputation for a specified 512-bit group, we can compute arbitrary discrete logs in that group in about a minute. We find that 82% of vulnerable servers use a single 512-bit group, allowing us to compromise connections to 7% of Alexa Top Million HTTPS sites. In response, major browsers are being changed to reject short groups.

We go on to consider Diffie-Hellman with 768- and 1024-bit

coded, or widely shared Diffie-Hellman parameters has the effect of dramatically reducing the cost of large-scale attacks, bringing some within range of feasibility today.

The current best technique for attacking Diffie-Hellman relies on compromising one of the private exponents (a, b) by computing the discrete log of the corresponding public value $(g^a \bmod p,\ g^b \bmod p)$. With state-of-the-art number field sieve algorithms, computing a single discrete log is more difficult than factoring an RSA modulus of the same size. However, an adversary who performs a large precomputation for a prime p can then quickly calculate arbitrary discrete logs in that group, amortizing the cost over all targets that share this parameter. Although this fact is well known among mathematical cryptographers, it seems to have been lost

Rivest–Shamir–Adleman Asymmetric Key Infrastructure

For every agent A, the key is split into a public key pk_A and secret key sk_A.

Pick primes p and q. Let $n = pq$ and $\varphi(n) = (p-1)(q-1)$.

Pick e such that $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.

Compute d as the multiplicative inverse of $e \mod \varphi(n)$.

Rivest–Shamir–Adleman Asymmetric Key Infrastructure

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Why does it exist? How do you compute it?

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Pick e such that $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.

Compute d as the multiplicative inverse of $e \mod \varphi(n)$.

$$ed \equiv 1 \mod \varphi(n)$$

Rivest–Shamir–Adleman Asymmetric Key Infrastructure

Public key is (e, n) and Secret Key is (d, n).

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To decrypt, compute $m^{ed} \mod n$.

Rivest–Shamir–Adleman Asymmetric Key Infrastructure

Public key is (e, n) and Secret Key is (d, n).

To encrypt a given message $m < n$, compute $m^e \mod n$.

To decrypt, compute $m^{ed} \mod n$.

$$m^{ed} \equiv m^{\varphi(n)+1} \equiv m \mod n$$

Rivest–Shamir–Adleman Asymmetric Key Infrastructure

$$m^{ed} \equiv m^{\varphi(n)+1} \equiv m \mod n$$

What are the assumptions? What are the limitations?

Is factoring NP-complete?